

# Circuits & Systems — Study Pack (MTE)

ECE2107 · B.Tech ECE Sem 3 · follows the professor's MTE syllabus (dated 10/9/26) · built from two sets of class notes · every answer checked by circuit simulation

## Priority order: if time is short, study in this order

Ranked by how much the professor stressed each topic: the number of board questions worked in class, and how many notes pages it took. There is **no past paper yet**, so this is an estimate. The one hard fact is the syllabus, and every item on it can come up.

| Rank | Study                                     | Why it ranks here                                                                                                            | Do first               | Rough time |
|------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------|------------|
| 1    | <b>06 Thévenin &amp; Norton</b>           | The professor's longest worked question (dependent source, nodal route); it appears in both notes sets                       | 06 §2–5, then mock B10 | 60 min     |
| 2    | <b>05 Superposition</b>                   | 2 board questions + the "limitation" question the professor highlighted                                                      | 05 §1–5, mock B7–B8    | 40 min     |
| 3    | <b>08 First order</b>                     | 4 board questions (switching at $0^-/0^+/\infty$ , $\tau$ , RC and RL); Laplace and test signals are separate syllabus items | 08 §1–4, then §6–7     | 60 min     |
| 4    | <b>07 Maximum power transfer</b>          | 2 board question sets, one of them six parts long; the derivation is a likely theory question                                | 07 §2–3, mock B12–B14  | 30 min     |
| 5    | <b>04 Nodal &amp; mesh</b>                | A syllabus item, and 06 depends on it (the professor solved $V_{Th}$ by nodal)                                               | 04 §1–4                | 40 min     |
| 6    | <b>09 Second-order RLC</b>                | A syllabus item with four named cases; no board question yet                                                                 | 09 §3–4, exam form     | 40 min     |
| 7    | <b>02 Dividers, power &amp; energy</b>    | 4 short board questions, all quick marks                                                                                     | 02 §1–4                | 25 min     |
| 8    | <b>03 KVL, KCL, source transformation</b> | Short; used inside every other method                                                                                        | 03 §1–4                | 20 min     |
| 9    | <b>01 Elements &amp; sources</b>          | Theory only (classification, dependent sources, lumped vs distributed)                                                       | 01 exam form           | 20 min     |
| last | <b>10 Mock paper</b>                      | Closed-book, after the rest                                                                                                  | all                    | 2–3 h      |

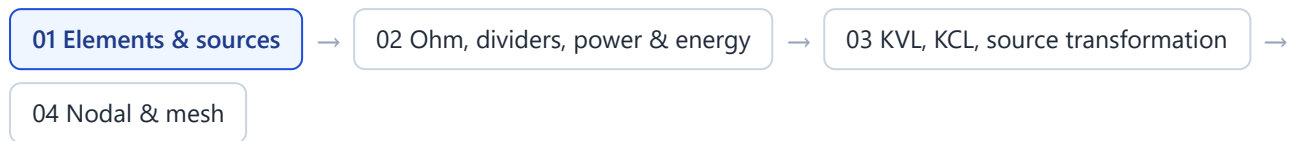
Two rules for using the ranking:

- **Short on time:** go down the list and do only the "Do first" column, then the matching mock-paper items.
- **Stuck on 06:** it builds on 05 (switching sources off) and 04 (nodal). Go back one step and don't skip ahead.

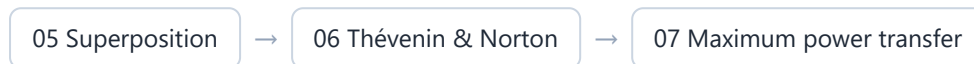
## The path through this pack

The order is the professor's MTE syllabus, item by item.

### Part I — elements, laws, analysis



### Part II — theorems (DC and AC)



### Part III — transient analysis



Each file needs the ones before it. 04 is KCL/KVL (03) written as a method. 06 uses 05's "switch a source off" rule, 04's nodal analysis and 01's dependent sources. 07 is built on 06. 08 uses 06 for the time constant; 09 uses 08's Laplace route.

| File | Syllabus item                       | What it teaches                                                                                                           | Board Qs               |
|------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------|
| 01   | 1 · elements (R, L, C, IDS, DS)     | active vs passive, independent and dependent sources (with devices), R, L, C laws, what can't jump, lumped vs distributed | —                      |
| 02   | 2 · laws                            | Ohm, voltage and current division (DC and AC), power sign convention, energy, ladders                                     | 4                      |
| 03   | 3 · KVL, KCL, source transformation | the two laws, power balance, source transformation                                                                        | 1 + 1<br>textbook      |
| 04   | 4 · nodal and mesh                  | both methods, with current sources and dependent sources                                                                  | 1 + 1<br>textbook      |
| 05   | 5 · superposition                   | method, the cases where it fails, AC sources at different frequencies                                                     | 2 + 1<br>textbook      |
| 06   | 5 · Thévenin & Norton               | with a dependent source; with AC impedances                                                                               | 1 + 1<br>textbook      |
| 07   | 5 · maximum power transfer          | $R_L = R_{Th}$ , efficiency, conjugate match (worked)                                                                     | 2 sets + 1<br>textbook |
| 08   | 6 · first order                     | $0^- / 0^+ / \infty$ , $\tau$ , the one formula, Laplace, test signals $\delta$ , $u$ , $r$                               | 4 + 2<br>textbook      |
| 09   | 7 · second order                    | series RLC: over-, critically, under-damped, undamped                                                                     | textbook               |
| 10   | all                                 | one paper: 12 theory prompts + 20 numericals                                                                              | 14 + 6<br>textbook     |

## Where the questions come from

A question that opens a section is **quoted from the professor's board** where one exists, and tagged with its notes page: **Class p.7** = first notes set (the 23-page scan), **Set 2 p.9** = second notes set (22 photos). Where the syllabus names a topic the class has not yet worked a question on, the pack uses a standard textbook-style problem, tagged **textbook**.

|           | Base                                                 | Status                                                                                                                                                                                 |
|-----------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1</b> | The professor's board questions (notes sets 1 and 2) | <b>14 questions</b> , all used                                                                                                                                                         |
| <b>P2</b> | Textbook problems                                    | <b>7</b> , written for syllabus items with no board question (KVL, nodal/mesh, AC theorems, Laplace, test signals, RLC). Values chosen so answers are clean; all checked by simulation |

Marks per question: **not known** (no hand-out, no past paper yet).

## How each file runs

| Part                      | What happens                                                                             |
|---------------------------|------------------------------------------------------------------------------------------|
| <b>Map</b>                | the file's pieces, in order                                                              |
| <b>Build</b>              | each step opens with a real question → you guess → the answer teaches the idea → a Check |
| <b>Exam form</b>          | what to reproduce in the exam: statements, procedures, the formula sheet                 |
| <b>Traps</b>              | the mistakes that cost marks                                                             |
| <b>Self-test, Answers</b> | answers are on the last page(s) of every file                                            |

### THE ONE RULE THAT MAKES THIS WORK

When you reach a **Guess first** box, write a guess on paper **before** reading on: one line, however wrong. A wrong guess that gets corrected sticks better than a right answer you only read.

Mark every Check and Self-test item:

- **clean** → revisit in 3 days
- **with struggle** → revisit tomorrow
- **wrong** → redo today, after the file

## Units and symbols used everywhere

- Uppercase **V, I** = DC (constant) values. Lowercase **v, i** = values that change with time. Complex values such as  $5 - j5$  V are **phasors**: amplitude and phase in one number.
- Resistances in **k $\Omega$**  with currents in **mA** give voltages in **V** directly:  $2 \text{ k}\Omega \times 3 \text{ mA} = 6 \text{ V}$ . Most questions use this.
- **R<sub>Th</sub>, V<sub>Th</sub>** = Thévenin resistance and voltage. **I<sub>N</sub>** = Norton current. **R<sub>L</sub>** = load resistance.
- **0<sup>-</sup>** = just before the switch moves. **0<sup>+</sup>** = just after.  **$\infty$**  = long after (steady state).

## Where the two notes sets disagree with the pack

| Topic                                  | Notes say                                        | Pack says                           | Why                                                                                                                                              |
|----------------------------------------|--------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Superposition, two 5 V sources (05 §5) | set 2: $0 + 0 = 0$ A, "LIMITATION"; set 1: 10 mA | 0 A by superposition, true 5 mA     | set 2 is the correct use of the rule and shows why it fails; the pack follows set 2                                                              |
| Thévenin with 2I (06 §2)               | set 2: $V_{Th} = 5.0025$ V                       | 20/3 V                              | set 2 reads 2I as $2 \Omega \times I$ ; the pack reads $2 \text{ k}\Omega \times I$ . Both are right for their reading. <b>Ask the professor</b> |
| RL switch that opens (08 §5)           | $I_R(0^+) = 0$ , $I_L(0^+) = 10$ mA              | cannot both hold in one series loop | <b>ask the professor</b> which path the current takes                                                                                            |

## How this pack was checked

- Facts: checked against the textbook-sourced circuits knowledge base (Hayt, Van Valkenburg, Sudhakar & Shyammohan) and Alexander & Sadiku.
- Numbers: every circuit in this pack was solved a second time by computer (nodal analysis and time simulation), and the answers here match. The script is `src/verify_mte.py`.
- Where the class notes had a slip of the pen, the pack uses the corrected version and says so in a Note.